IMU-Based Head Tracking for Camera Control in a Virtual Driving Environment

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1 Background

Computer graphics and interaction systems often rely on intuitive input methods to control virtual environments. In applications such as driving simulators and virtual reality, head tracking is commonly used to control the user's field of view (FOV), enabling more immersive interaction. The inspiration for this project comes from the scenario when I was playing the Euro Truck Simulator 2, if the head tracking feature is enabled, it normally requires the player to invest relatively expensive head and eye tracking devices [1].
In this project, I explore the use of a low-cost inertial measurement unit (IMU), specifically the MPU6050 sensor[2], along with the Arduino Nano to capture head motion and map it to camera movement in a virtual environment. The MPU6050 provides accelerometer and gyroscope data, which can be used to estimate orientation in terms of roll, pitch and yaw.
Another main reason for me to select this project is due to my previous background in electronics and embedded system, I would like to use this project as a chance to explore the interaction between reality and virtual world.

2 Problem Statement

The main goal of this project is to create an interactive system where head movements control the camera orientation inside a virtual driving cockpit.
There are some challenging part in this project:

- Raw IMU data is noisy and unstable, especially for gyroscope-based orientation.
- Yaw estimation tends to drift over time due to sensor limitations.
- Mapping real-world motion to virtual camera movement must feel smooth and responsive.
- Real-time performance must be maintained while processing sensor data and rendering graphics.
- Creating the realistic virtual driving cockpit is time-consuming.

- It is planned to create the virtual environment within the processing, it could be challenging to integrate the sensor input with the processing.

3 Implementation

The system consists of both hardware and software components.

3.1 Hardware

- MPU6050 IMU sensor mounted on the user's head
- Arduino Nano (or potentially using UNO for testing) for reading sensor data

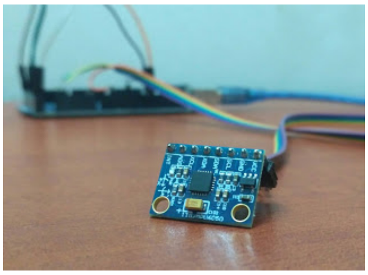

Figure 1: Arduino and MPU5060 sensor.

3.2 Software

- Arduino IDE for sensor data acquisition and testing
- Processing for creating the virtual cockpit and implementing the real-time camera position adjustment from serial inputs
- If more time allowed, self-learning and designing the car interior in Blender then implemented into processing could achieve a better visualization

The Fig. 2 shows the example project I have found on YouTube that was created by Arduino developer DIY Engineers using the same MPU5060 sensor that connected to a 3-D printed plane-like plate to realize the digital twin-like system[3].

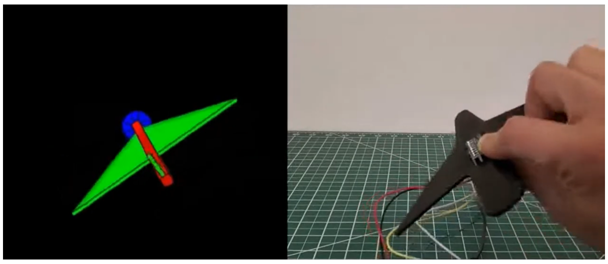

Figure 2: MPU5060-based digital twin example project.

3.3 Graphics Implementation

A simple 3D virtual environment representing a car cockpit will be created in Processing. The camera will be placed at the driver's position and will rotate according to the user's head movement.
The following transformations will be applied:

- Yaw (left/right head movement) → rotation around Y-axis
- Pitch (up/down movement) → rotation around X-axis

3.4 Filtering

To improve stability, a simple complementary filter or smoothing method will be applied to combine accelerometer and gyroscope data. The goal is to reduce noise while maintaining responsiveness.

4 Specification Ideas

The project will include the following components:

4.1 Interaction

- Real-time head tracking using IMU sensor
- Mapping physical motion to virtual camera movement

4.2 Visualization

- 3D cockpit scene (simplified car interior created using Processing or Blender)
- Camera-based navigation controlled by head motion

4.3 Evaluation Measures

- Smoothness of motion (qualitative observation)
- Responsiveness (latency between motion and update)
- Stability (comparison between raw and filtered data)

4.4 Comparison

- Raw sensor data vs filtered data
- Different smoothing parameters

5 Expected Results

The expected outcome is a working prototype where users can control the camera view inside a virtual driving environment using head movement.
It is expected that filtering techniques will significantly improve the stability and usability of the system. The project will demonstrate how simple sensor processing methods can enhance interaction in computer graphics applications.

6 Challenges

The project involves both hardware and software integration, and based on the current project timeline the time scheme is limited to achieve the best system performance.
The delivery and cost of the hardware could potentially slow down the project progress. Designing and tuning the filter of the input data could also be the most time consuming process.

7 Target Grade

The initial goal of this project is to achieve a grade in the C/D range by successfully implementing the system using existing resources.
If time allows, additional features such as improved filtering, detailed design of virtual cockpit, or extended interaction will be added to aim for a higher grade (B).

8 References

References

[1] Tobii Gaming, "Euro Truck Simulator 2 - Eye and Head Tracking," Available: `https://gaming.tobii.com/games/euro-truck-simulator-2/`. Accessed: Apr. 20, 2026.

[2] Rucksikaa Raajkumar, "MPU-6050 Sensor Module: A Friendly Introduction," Arduino Project Hub, Feb. 2, 2021. Available: `https://projecthub.arduino.cc/RucksikaaR/mpu-6050-sensor-module-a-friendly-introduction-7b4d4b`. Accessed: Apr. 20, 2026.

[3] DIY Engineers, "MPU6050 with Arudino - GY-521," May. 6, 2021. Available: `https://youtu.be/oqmbVeRlLgg?si=XMoeTjbOm7AYVMPK` Accessed: Apr. 20, 2026.